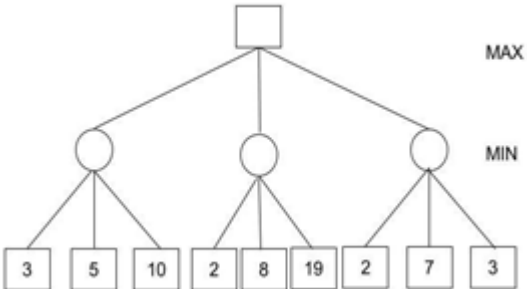


Course Code:B23ADOE05					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
PRINCIPLES OF ARTIFICIAL INTELLIGENCE					
Offered by AIDS					
(offered to CE, ECE, EEE & ME)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	What is the goal of an intelligent agent?	1	2	2
	b).	What are some common applications of AI?	1	2	2
	c).	What challenges are faced in Hill climbing Algorithm?	2	2	2
	d).	What is a general problem-solving method (GPS)?	2	2	2
	e).	What is propositional calculus and operators used in propositional calculus?	3	2	2
	f).	List the any two Axioms in Axiomatic system	3	1	2
	g).	Name any two types of commonly used membership functions draw their diagrams.	4	2	2
	h).	Give a small example of knowledge representation using semantic network in everyday life.	4	2	2
	i).	Define Bayes' Theorem. How does it handle uncertainty?	5	1	2
	j).	What are expert systems and its applications.	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	What is the role of a "performance measure" in different types of agents?	1	1	10
		OR			
3.	a).	Demonstrate TIC TAC TOE approach 3 with example?	1	3	10
		UNIT-2			
4.	a).	You have two jugs, one with a capacity of 5 liters and the other with a capacity of 3 liters. How can you measure exactly 4 liters of water?	1	3	10
		OR			
5.	a).	Given the following graph with edge weights and heuristic values,	2	3	10

		use the A* algorithm to find the shortest path from node A to node Z.			
		UNIT-3			
6.	a).	Explain alpha beta pruning on below example? 	3	3	10
		OR			
7.	a).	Show that following is a valid argument: "If I wake up early or I have an exam, then I will drink coffee and study."	3	3	10
		UNIT-4			
8.	a).	Explain extended semantic networks for knowledge representation.?	4	2	5
	b).	How do you represent visiting library in form of Script? Explain in detail?	4	2	5
		OR			
9.	a).	Write about Conceptual dependency theory and how it is used for Knowledge Representation?	4	3	10
		UNIT-5			
10.	a).	Explain the main components of a planning system?	5	2	5
	b).	Draw the architecture of expert system in detail?	5	3	5
		OR			
11.	a).	Differentiate Expert systems vs Traditional system?	5	2	5
	b).	Describe working mechanism of Goal stack planning and hierarchal planning?	5	3	5

COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23AMOE06					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
DEEP LEARNING					
Offered by AIML					
(offered to CE, ECE, EEE & ME)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Differentiate Supervised and Unsupervised Learning	1	2	2
	b).	Explain the difference between Bias and Variance in machine learning models?	1	2	2
	c).	Explain the purpose of Dropout in Deep Neural Networks.	2	2	2
	d).	Describe how L2 regularization helps prevent overfitting in neural networks.	2	2	2
	e).	Describe the role of the forget gate in an LSTM cell?	3	2	2
	f).	Explain the role of Batch Normalization in Convolutional Neural Networks.	3	2	2
	g).	Explain the role of learning rate in gradient descent?	4	2	2
	h).	Outline how Adam optimizer is preferred over traditional gradient descent methods?	4	2	2
	i).	Summarize the concept of Transfer Learning in Deep Learning.	5	2	2
	j).	Describe how LSTM can be used for Sentiment Analysis.	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Describe the purpose of Cross-Validation. Explain how is it useful in model evaluation and selection?	1	2	5
	b).	Describe the purpose of Hyper parameter Tuning. Describe how Hyper Parameters are different from model parameters?	1	2	5
		OR			
3.	a).	Differentiate Overfitting and Under fitting in machine learning models. Explain how do they affect model performance?	1	2	5
	b).	Define Dimensionality Reduction? Explain any one technique used for it and its advantages in machine learning.	1	2	5

		UNIT-2			
4.	a).	For weight vector $w = [0.5, -0.4, 0.3]$ and base loss $L = 0.25$, compute regularized loss using (a) L1 with $\lambda = 0.05$ and (b) L2 with $\lambda = 0.05$. Find which penalty is stronger.	2	3	5
	b).	Illustrate Various Activation Functions with examples	2	2	5
		OR			
5.	a).	Illustrate how Backpropagation is applied in a Multilayer Perceptron with one hidden layer using a simple numerical example.	2	3	5
	b).	Explain the concept of error functions in deep learning. How do they influence the training process of a neural network?	2	2	5
		UNIT-3			
6.	a).	Illustrate Recurrent Neural Networks	3	2	5
	b).	Explain about Long Short-Term Memory	3	2	5
		OR			
7.	a).	Explain the architecture of a Convolutional Neural Network. Describe the role of convolution, pooling, and batch normalization layers.	3	2	5
	b).	Describe the purpose of Normalization in CNNs. How does Batch Normalization help improve training?	3	2	5
		UNIT-4			
8.	a).	Describe about Autoencoders and Explain how they are used	4	2	5
	b).	Explain about stochastic gradient descent	4	2	5
		OR			
9.	a).	Describe Adam Algorithm	4	2	5
	b).	Discuss in detail about Denoising Auto encoders with a suitable example.	4	2	5
		UNIT-5			
10.	a).	Illustrate Alexnet architecture	5	2	5
	b).	Explain how to improve the performance of a model with Transfer learning	5	3	5
		OR			
11.	a).	Illustrate Res Net architecture	5	2	5
	b).	Explain the steps involved in sentiment analysis using LSTM with an example	5	3	5

COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23AMOE07					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
OBJECT ORIENTED PROGRAMMING THROUGH JAVA					
Offered by AIML					
(offered to CE, ECE, EEE & ME)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	What are the advantages of OOP?	1	1	2
	b).	Describe a scenario where using this keyword is necessary in a Java class.	1	2	2
	c).	What is type conversion in Java, and why is it necessary?	2	1	2
	d).	Explain String Constant Pool	2	2	2
	e).	What are the uses of Super Keyword?	3	1	2
	f).	What is the use of the final keyword in Java?	3	1	2
	g).	List the four access modifiers in Java.	4	1	2
	h).	What are the differences between throw and throws keyword?	4	1	2
	i).	What are thread priorities in Java, and how do they affect the execution of threads?	5	1	2
	j).	What is the difference between FileInputStream and FileReader?	5	1	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Explain Features of JAVA.	1	2	5
	b).	Write a program to accept two integers as command line arguments and print the sum of the two numbers.	1	3	5
		OR			
3.	a).	Explain Method Overloading with an example.	1	2	5
	b).	Explain different types of Constructors with suitable examples.	1	2	5
		UNIT-2			
4.	a).	Write a program to reverse the elements of a given 2*2 array. Four integer numbers need to be passed as Command Line arguments.	2	3	5

	b).	Explain any five commonly used methods in the String class with examples.	2	2	5
		OR			
5.	a).	What is a two-dimensional array? How is it declared and initialized in Java? Provide a sample program.	2	2	5
	b).	Differentiate String and StringBuffer class.	2	3	5
		UNIT-3			
6.	a).	Explain different types of inheritance with an example.	3	2	5
	b).	Differentiate Interface and Abstract class.	3	3	5
		OR			
7.	a).	Create an interface Vehicle with a default method message () that returns nothing and prints "Inside Vehicle". Create an interface FourWheeler with a default method message () that returns nothing and prints "Inside FourWheeler". Create a class Car implementing these two interfaces. In this class, Override the message () method and call the message () method of the Vehicle interface using super keyword. Instantiate the class, call the message method and print the output.	3	3	5
	b).	Explain method overriding with suitable example.	3	2	5
		UNIT-4			
8.	a).	What are sub-packages in Java? How do you create and access them? Give a code example.	4	2	5
	b).	Write a program that take an input String from user and parse it to integer, if it is not a number it will throw number format exception Catch it and print "Entered input is not a valid format for an integer." or else print the square of that number.	4	3	5
		OR			
9.	a).	Explain Multiple Catch Statements with an example.	4	2	5
	b).	Create a package called test package. Define a class called foundation inside the test package. Inside the class, you need to define 4 integer variables: var1 with private access modifier, var2 with default access modifier, var3 with protected access modifier and var4 with public access modifier. Import this class and packages in another class. Try to access all 4 variables of the foundation class and see what variables are accessible and what are not accessible.	4	3	5
		UNIT-5			
10.	a).	Explain Thread Life Cycle with a neat diagram.	5	2	5

	b).	Write a program to copy contents of one file to another file.	5	3	5
		OR			
11.	a).	Write a java program to implement Thread Synchronization.	5	3	5
	b).	Explain the purpose of the Reader and Writer classes in Java. How do they differ from InputStream and OutputStream?	5	2	5

COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks



SRKR
ENGINEERING COLLEGE
AUTONOMOUS

Course Code:B23CEO05					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
SUSTAINABILITY IN ENGINEERING PRACTICES					
Offered by CE					
(offered to AIDS, AIML, CIC, CSBS, CSE, CSG, CSIT, ECE, EEE, IT & ME)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define sustainable development.	1	1	2
	b).	What is ecological footprint?	1	1	2
	c).	Define resource degradation.	2	1	2
	d).	What is climate change?	2	1	2
	e).	What is Environmental Management System (EMS)?	3	1	2
	f).	Define Life Cycle Assessment (LCA).	3	1	2
	g).	What is green building?	4	1	2
	h).	Define GRIHA rating.	4	1	2
	i).	What are renewable energy resources?	5	1	2
	j).	Define biomass energy.	5	1	2
5 x 10 =50Marks					
		UNIT-1			
2.	a).	Explain the principles of sustainable development and sustainability models.	1	2	10
		OR			
3.	a).	Describe environmental ethics and discuss the role of policies and protocols in sustainable development.	1	2	10
		UNIT-2			
4.	a).	Explain local environmental issues such as solid waste, water resources, and land degradation.	2	2	10
		OR			
5.	a).	Discuss global environmental issues including climate change, ozone depletion, and carbon footprint.	2	2	10

		UNIT-3			
6.	a).	Explain Environmental Management System (EMS) and ISO 14000 standards.	3	2	10
		OR			
7.	a).	Illustrate Life Cycle Assessment (LCA) and Environmental Impact Assessment (EIA) with examples.	3	3	10
		UNIT-4			
8.	a).	Explain concepts of green buildings and rating systems such as GRIHA and LEED.	4	2	10
		OR			
9.	a).	Apply sustainable practices in urban development and industrial sustainability.	4	3	10
		UNIT-5			
10.	a).	Describe different renewable energy resources and their applications.	5	2	10
		OR			
11.	a).	Explain green technologies such as green buildings, green transportation, and green computing.	5	2	10

COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Estd. 1980

AUTONOMOUS

Course Code: B23CEOE06					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
SOLID WASTE MANGEMENT					
Offered by CE					
(offered to AIDS, AIML, CIC, CSBS, CSE, CSG, CSIT, ECE, EEE, IT & ME)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
					10 x 2 = 20 Marks
			CO	KL	M
1.	a).	Define solid waste and give examples.	1	1	2
	b).	List functional elements of solid waste management.	1	1	2
	c).	Explain is moisture content of waste?	2	2	2
	d).	Explain route optimization?	2	2	2
	e).	Explain what is metal separation.	3	2	2
	f).	Define anaerobic digestion.	3	1	2
	g).	What is landfill remediation?	4	1	2
	h).	Explain leachate?	4	2	2
	i).	List characteristics of hazardous waste.	5	1	2
	j).	What is incineration of hazardous waste?	5	2	2
					5 x 10 = 50 Marks
		UNIT-1	CO	KL	M
2.	a).	Explain sources, types and characteristics of solid waste in detail.	1	2	5
	b).	Describe methods of sampling and characterization of municipal solid waste.	1	2	5
		OR			
3.	a).	Explain onsite storage and handling practices of solid waste.	1	2	5
	b).	Discuss future changes in waste composition and roles of legislation.	1	3	5
		UNIT-2			
4.	a).	Explain different types of solid waste collection systems.	2	2	5
	b).	Describe transfer stations, their need.	2	2	5
		OR			
5.	a).	Explain transportation methods of solid waste and their importance.	2	2	5
	b).	Discuss recycling and reuse potential of solid waste as a resource.	2	3	5

		UNIT-3			
6.	a).	Explain material recovery facilities and source reduction techniques.	3	2	5
	b).	Describe composting and anaerobic digestion processes.	3	2	5
		OR			
7.	a).	Explain incineration process and its limitations.	3	2	5
	b).	Discuss energy recovery methods from solid waste.	3	3	5
		UNIT-4			
8.	a).	Explain site selection criteria for landfills.	4	2	5
	b).	Describe landfill operation methods.	4	3	5
		OR			
9.	a).	Explain leachate collection system and its importance.	4	2	5
	b).	Discuss landfill closure and environmental protection methods.	4	2	5
		UNIT-5			
10.	a).	Explain sources of hazardous waste.	5	2	5
	b).	Describe hazardous waste handling, storage and transportation.	5	2	5
		OR			
11.	a).	Explain treatment methods of hazardous waste (physical & chemical).	5	2	5
	b).	Discuss e-waste management and regulations.	5	3	5

COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Course Code:B23CBOE04					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
DIGITAL MARKETING					
Offered by CSBS					
(offered to CE, ECE, EEE & ME)					
Time:3Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Discuss the significance of digital marketing.	1	2	2
	b).	Describe the development of internet business models after the Dotcom era.	1	2	2
	c).	Explain the concept of CPL.	2	2	2
	d).	Define the concept of display advertising.	2	2	2
	e).	Summarize the concept of search engine advertising.	3	2	2
	f).	Explain key elements of Search Engine Optimization (SEO).	3	2	2
	g).	Explain Goal Setting in social media marketing.	4	2	2
	h).	Explain how Instagram and Snapchat are used for marketing.	4	2	2
	i).	Explain the impact of IoT on marketing.	5	2	2
	j).	Explain the concept of SoLoMo marketing.	5	2	2
5 x 10 = 50 Marks					
		UNIT-1	CO	KL	M
2.	a).	Describe the evolution of digital marketing from traditional to modern marketing.	1	2	5
	b).	Discuss the role of the internet in the growth of digital marketing.	1	2	5
		OR			
3.	a).	Explain the importance of the Dotcom Era in digital marketing development.	1	2	5
	b).	Explain various applications of digital marketing in business.	1	2	5
		UNIT-2			
4.	a).	Describe different types of display advertising media.	2	2	5
	b).	Discuss the role of digital metrics in display advertising.	2	2	5

		OR			
5.	a).	Explain the CPC and CPM buying models.	2	2	5
	b).	Summarize fixed cost and sponsorship-based advertising models.	2	2	5
		UNIT-3			
6.	a).	Explain how search engines work.	3	2	5
	b).	Discuss the importance of paying for search advertising.	3	2	5
		OR			
7.	a).	Describe different ad placements in search engines.	3	2	5
	b).	Discuss the importance of SEO in digital marketing.	3	2	5
		UNIT-4			
8.	a).	Explain the concept of social media marketing.	4	2	5
	b).	Explain Facebook marketing and its features.	4	2	5
		OR			
9.	a).	Describe the steps involved in building a successful social media strategy.	4	2	5
	b).	Describe LinkedIn marketing and its professional advantages.	4	2	5
		UNIT-5			
10.	a).	Apply the concept of digital marketing spend to analyze the Indian market.	5	3	5
	b).	Demonstrate how Big Data can be used in digital marketing strategies.	5	3	5
		OR			
11.	a).	Apply digital marketing strategies for B2B segments.	5	3	5
	b).	Apply the concept of SoLoMo in a hyperlocal marketing scenario.	5	3	5

COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23CSOE04					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
OBJECT ORIENTED PROGRAMMING THROUGH JAVA					
Offered by CSE					
(offered to CE, ECE, EEE & ME)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	What are the advantages of OOP?	1	1	2
	b).	Describe a scenario where using this keyword is necessary in a Java class.	1	2	2
	c).	What is type conversion in Java, and why is it necessary?	2	1	2
	d).	Explain String Constant Pool	2	2	2
	e).	What are the uses of Super Keyword?	3	1	2
	f).	What is the use of the final keyword in Java?	3	1	2
	g).	List the four access modifiers in Java.	4	1	2
	h).	What are the differences between throw and throws keyword?	4	1	2
	i).	What are thread priorities in Java, and how do they affect the execution of threads?	5	1	2
	j).	What is the difference between FileInputStream and FileReader?	5	1	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Explain Features of JAVA.	1	2	5
	b).	Write a program to accept two integers as command line arguments and print the sum of the two numbers.	1	3	5
		OR			
3.	a).	Explain Method Overloading with an example.	1	2	5
	b).	Explain different types of Constructors with suitable examples.	1	2	5
		UNIT-2			
4.	a).	Write a program to reverse the elements of a given 2*2 array. Four integer numbers need to be passed as Command Line arguments.	2	3	5
	b).	Explain any five commonly used methods in the String class with	2	2	5

		examples.			
		OR			
5.	a).	What is a two-dimensional array? How is it declared and initialized in Java? Provide a sample program.	2	2	5
	b).	Differentiate String and StringBuffer class.	2	3	5
		UNIT-3			
6.	a).	Explain different types of inheritance with an example.	3	2	5
	b).	Differentiate Interface and Abstract class.	3	3	5
		OR			
7.	a).	Create an interface Vehicle with a default method message () that returns nothing and prints "Inside Vehicle". Create an interface FourWheeler with a default method message () that returns nothing and prints "Inside FourWheeler". Create a class Car implementing these two interfaces. In this class, Override the message () method and call the message () method of the Vehicle interface using super keyword. Instantiate the class, call the message method and print the output.	3	3	5
	b).	Explain method overriding with suitable example.	3	2	5
		UNIT-4			
8.	a).	What are sub-packages in Java? How do you create and access them? Give a code example.	4	2	5
	b).	Write a program that take an input String from user and parse it to integer, if it is not a number it will throw number format exception Catch it and print "Entered input is not a valid format for an integer." or else print the square of that number.	4	3	5
		OR			
9.	a).	Explain Multiple Catch Statements with an example.	4	2	5
	b).	Create a package called test package. Define a class called foundation inside the test package. Inside the class, you need to define 4 integer variables: var1 with private access modifier, var2 with default access modifier, var3 with protected access modifier and var4 with public access modifier. Import this class and packages in another class. Try to access all 4 variables of the foundation class and see what variables are accessible and what are not accessible.	4	3	5
		UNIT-5			
10.	a).	Explain Thread Life Cycle with a neat diagram.	5	2	5
	b).	Write a program to copy contents of one file to another file.	5	3	5

		OR			
11.	a).	Write a java program to implement Thread Synchronization.	5	3	5
	b).	Explain the purpose of the Reader and Writer classes in Java. How do they differ from InputStream and OutputStream?	5	2	5

COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks



SRKR
ENGINEERING COLLEGE
AUTONOMOUS

Course Code: B23CIOE05					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
INTRODUCTION TO CRYPTOGRAPHY AND NETWORK SECURITY					
Offered by CIC					
(offered to CE, ECE, EEE & ME)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define Phishing and give an example.	1	1	2
	b).	What is a block cipher? How does it differ from stream cipher?	1	2	2
	c).	Write any two advantages of Advanced Encryption Standard (AES).	2	1	2
	d).	Explain the differences between RC4 and RC5.	2	2	2
	e).	What is a cryptographic hash function? Mention its use.	3	2	2
	f).	Write short notes on HMAC.	3	1	2
	g).	Define Kerberos and state its purpose.	4	1	2
	h).	What is an Authentication Header (AH) in IP Security?	4	2	2
	i).	Write any two characteristics of firewalls.	5	1	2
	j).	What are the differences between SSL and TLS?	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Explain different types of Security Attacks and Security Services.	1	2	5
	b).	Illustrate the working of Substitution and Transposition techniques with examples.	1	2	5
		OR			
3.	a).	Describe Block Cipher Modes of Operation.	1	2	5
	b).	Explain SQL Injection and Web-based attacks.	1	2	5
		UNIT-2			
4.	a).	Explain the architecture and working of AES algorithm.	2	2	5
	b).	Perform Encryption and Decryption using RSA algorithm for p=17, q=11, e=7, M=88.	2	3	5
		OR			
5.	a).	Calculate the secret key shared between User A and User B using Diffie	2	3	5

		Hellman Key exchange algorithm for the following: $q=97$, $a=5$, the private keys $X_A = 36$, $X_B = 58$.			
	b).	Illustrate Elliptic Curve Cryptography (ECC)	2	2	5
		UNIT-3			
6.	a).	Apply MD5 algorithm to generate hash value with an example.	3	3	5
	b).	Explain HMAC and CMAC in detail.	3	2	5
		OR			
7.	a).	Use DSA to sign and verify signature with an example.	3	3	5
	b).	Discuss Elgamal Digital Signature Scheme.	3	2	5
		UNIT-4			
8.	a).	Explain the working of Kerberos authentication protocol.	4	2	5
	b).	Describe the architecture and functionality of IP Security (IPSec).	4	2	5
		OR			
9.	a).	Discuss the PGP Protocol for Email Security.	4	2	5
	b).	What is S/MIME? Describe how it ensures secure email communication.	4	2	5
		UNIT-5			
10.	a).	What are the Web Security Requirements? Explain in the context of SSL.	5	2	5
	b).	Explain various types of firewalls with diagrams.	5	2	5
		OR			
11.	a).	Describe the architecture and working of TLS protocol.	5	2	5
	b).	Explain the configuration of firewalls in secure communication.	5	2	5

COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23CIOE06					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
COMPUTER NETWORKS					
CIC					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Describe the key features of the OSI Reference Model.	1	2	2
	b).	Explain the role of network protocols in communication between devices.	1	2	2
	c).	Describe the need for error control mechanisms in communication systems.	2	2	2
	d).	Explain framing in the data link layer.	2	2	2
	e).	Describe how routing works in a network.	3	2	2
	f).	Explain IPv4 addressing.	3	2	2
	g).	Explain the services provided by the transport layer to upper layers.	4	2	2
	h).	Explain the role of congestion control in TCP.	4	2	2
	i).	Describe the purpose of content delivery networks.	5	2	2
	j).	Explain the concept of peer-to-peer networks.	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Explain various types of computer networks.	1	2	5
	b).	Describe the concept of protocol layering, including its advantages and how it simplifies network design.	1	2	5
		OR			
3.	a).	Explain the TCP/IP Reference Model and compare it with the OSI model.	1	2	5
	b).	Describe the relationship between services and protocols, including service primitives and their role in communication.	1	2	5
		UNIT-2			
4.	a).	Explain the various types of guided transmission media.	2	2	5
	b).	Apply the CRC technique to the data word 1101011011 using the generator polynomial 10011 and determine the transmitted codeword.	2	3	5
		OR			

5.	a).	Illustrate how elementary data link protocols can be used to ensure reliable communication over a noisy channel.	2	2	5
	b).	Demonstrate the working of Ethernet.	2	3	5
		UNIT-3			
6.	a).	An organization has been assigned the IP block 200.100.50.0/26. (a) Divide this block into 4 equal subnets. (b) For each subnet, write: 1. Subnet Mask 2. Network Address 3. First Host Address 4. Last Host Address 5. Broadcast Address	3	3	10
		OR			
7.	a).	Demonstrate Dijkstra's routing algorithm to determine the shortest path between a given source and destination node in a network graph.	3	3	5
	b).	Demonstrate how multicasting can be applied to efficiently deliver data to multiple recipients in a network.	3	3	5
		UNIT-4			
8.	a).	Explain how flow control is achieved in the transport layer.	4	2	5
	b).	Demonstrate how socket programming can be used to establish communication between a client and a server.	4	3	5
		OR			
9.	a).	Explain the differences between UDP and TCP in terms of connection type, reliability and speed.	4	2	5
	b).	A TCP connection is established between Host A and Host B. The initial sequence number of A is 1000, and B is 5000. Show the three-way handshake sequence numbers and acknowledgment numbers. Explain how TCP ensures reliable connection establishment.	4	3	5
		UNIT-5			
10.	a).	Demonstrate the working of electronic mail architecture to illustrate how an email is sent and received between two users.	5	3	10
		OR			
11.	a).	Apply the concept of dynamic web applications to design a simple web-based system.	5	3	5
	b).	Demonstrate how HTTP/HTTPS protocols are used to access web pages securely.	5	3	5

COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Course Code:B23ECOE03					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
OPTICAL COMMUNICATION					
Offered by ECE					
(Offered to AIDS, AIML, CE, CIC, CSBS, CSE, CSG, CSIT, EEE, IT &ME)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer ONE Question from EACH UNIT					
All questions carry equal marks					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	List out advantages of optical fiber.	1	2	2
	b).	Calculate the number of modes propagating in a graded index fiber at 880nm with 20µm core radius and $n_1=1.40$ and $n_2=1.46$.	1	3	2
	c).	What is dispersion in an optical fiber?	2	2	2
	d).	List the various losses in the optical fibers?	2	1	2
	e).	Illustrate in brief about Chemical vapour deposition.	3	1	2
	f).	List two properties of plastic optical fibers.	3	2	2
	g).	Define quantum efficiency of a LASER diode.	4	2	2
	h).	Draw the radiation pattern of an edge emitting LED.	4	1	2
	i).	What is system rise time and what is its significance in an optical communication link.	5	2	2
	j).	Name the types of stations used in FDDI network.	5	1	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Discuss the properties and characteristics of optical fibers.	1	3	5
	b).	Determine the core radius necessary for a single mode operation at 0.02µm for a step index fiber having $n_1= 1.45$ and $n_2=1.43$.	1	2	5
		OR			
3.	a).	Compare and contrast a step-index fiber with a graded index fiber. What is the advantage of using a graded index core in a fiber? Explain how this advantage is obtained.	1	2	5
	b).	With a neat block diagram, explain the functional blocks of an optical communication system.	1	3	5
		UNIT-2			
4.	a)	Explain how energy is lost from (i) leaky mode and (ii) fiber at sharp bend.	2	3	5

	b).	Explain the different types of signal distortions in optical fibers.	2	2	5
		OR			
5.	a).	A single mode step index fiber has a core diameter of 7 μm and core refractive index of 1.49. Estimate the shortest wavelength of light which allows single mode operation when the relative refractive difference for the fiber is 1.0%.	2	3	5
	b).	Describe the PCVD fabrication process for Optical fiber communication	2	3	5
		UNIT-3			
6.	a).	What are splices? Explain different types of splicing with neat sketches.	3	2	5
	b).	Discuss about inherent connection problem when making joints.	3	3	5
		OR			
7.	a).	Discuss about different Opto couplers and connectors.	3	3	5
	b).	Discuss in detail the fabrication and need for couplers.	3	3	5
		UNIT-4			
8.	a).	What is a hetero junction structures? Describe the two basic LED configurations that are suitable for optical fibers with neat schematic diagrams.	4	3	5
	b).	What are the various types of photo detectors used in optical communications? Compare their performances.	4	2	5
		OR			
9.	a).	Derive expressions for 3 decibels electrical and optical band width of an LED in terms of its effective life time.	4	3	5
	b).	Compare LASER and LED as optical sources.	4	2	5
		UNIT-5			
10.	a).	What is system rise time? Explain how does it limit an optical fiber communication link.	5	3	5
	b).	What are the various elements of an optical transmission link? Explain each element in detail.	5	3	5
		OR			
11.	a).	What are the types of pre- amplifiers used in optical receivers? Explain them briefly.	5	3	5
	b).	With the help of a schematic diagram explain the operational principles of WDM.	5	3	5

COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Course Code:B23ECOE04					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
FUNDAMENTALS OF VLSI DESIGN					
Offered by ECE					
(Offered to AIDS, AIML, CE, CIC, CSBS, CSE, CSG, CSIT, EEE, IT &ME)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer ONE Question from EACH UNIT					
All questions carry equal marks					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	What is the relation between I_{DS} and V_{DS} in MOSFET?	1	2	2
	b).	What is latch-up in CMOS circuits?	1	2	2
	c).	What is the purpose of using stick diagrams?	2	2	2
	d).	Draw the layout for the CMOS Inverter?	2	3	2
	e).	Define propagation delay in a CMOS inverter.	3	1	2
	f).	What is scaling and limitations of scaling?	3	1	2
	g).	What is the significance of LUTs in FPGA?	4	1	2
	h).	Define Level Sensitive Scan Design (LSSD).	4	2	2
	i).	List two static properties of Complementary CMOS.	4	2	2
	j).	What is the basic principle of Dynamic CMOS logic?	4	1	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Explain the CMOS fabrication steps with neat diagram?	1	3	6
	b).	Compare CMOS, Bi-CMOS and Bipolar Technologies?	1	3	4
		OR			
3.		Illustrate the different forms of pull-up transistors in MOS inverter circuits. Also provide the voltage transfer characteristics for each of them?	1	3	10
		UNIT-2			
4.		Draw the Stick Diagram of 2 input NOR gate using CMOS logic?	2	3	10
		OR			
5.		Explain CMOS design rule for wires?	2	2	10
		UNIT-3			

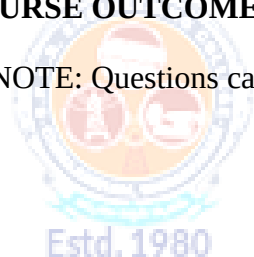
6.		Explain the concept of sheet resistance (R_s) and how it is applied to MOS transistors and inverters?	3	3	10
		OR			
7.		What are inverter delays? Explain the factors that contribute to propagation delay in CMOS inverters?	3	2	10
		UNIT-4			
8.		What are Scaling models and derive all scaling factors for device parameters.	4	2	10
		OR			
9.		Explain various limitations of scaling.	4	2	10
		UNIT-5			
10.		Explain the working principle of Complementary CMOS logic. What are its key static properties?	4	3	10
		OR			
11.		Design and Explain Multiplexer based Latches?	4	3	10

COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks



SRKR
ENGINEERING COLLEGE
AUTONOMOUS

Course Code: B23ECOE05					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
ELECTRONIC MEASUREMENTS AND INSTRUMENTATION					
Offered by ECE					
(Offered to AIDS, AIML, CE, CIC, CSBS, CSE, CSG, CSIT, IT &ME)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer ONE Question from EACH UNIT					
All questions carry equal marks					
Assume suitable data if necessary					
					10 x 2 = 20 Marks
			CO	KL	M
1.	a).	Define Accuracy and Precision.	1	1	2
	b).	Compare Dual Beam and Dual Trace CROs.	2	4	2
	c).	Illustrate the basic block diagram of a measurement system.	1	2	2
	d).	Use Maxwell's Bridge to find an unknown inductance formula.	3	3	2
	e).	List the requirements of a good Transducer.	5	1	2
	f).	Contrast Static and Dynamic characteristics.	1	4	2
	g).	Demonstrate how a Function Generator creates a square wave.	4	3	2
	h).	Classify transducers.	5	2	2
	i).	Define the term Sensitivity in voltmeters.	1	1	2
	j).	What is the primary purpose of signal generator.	2	1	2
					5 x 10 = 50 Marks
		UNIT-1			
2.		Explain the working of a True RMS responding voltmeter.	1	2	10
		OR			
3.		Analyze the various sources of errors in measurement systems.	1	4	10
		UNIT-2			
4.		Describe the Horizontal Deflection System of a CRO in detail.	2	2	10
		OR			
5.		Examine the operation of a Sampling Oscilloscope for high-frequency signals.	2	3	10
		UNIT-3			
6.		Derive the balance equation for a Schering Bridge.	3	3	10
		OR			

7.		Compare and contrast Maxwell's and Anderson bridges.	3	4	10
		UNIT-4			
8.		Outline the construction of a Basic Standard Signal Generator.	4	1	10
		OR			
9.		Demonstrate the working of a Square and Pulse Generator.	4	3	10
		UNIT-5			
10.		Explain the principle of Strain Gauges and derive the gauge factor.	5	2	10
		OR			
11.		Analyze the performance of a Linear Variable Differential Transducer (LVDT).	5	4	10

COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks



SRKR
ENGINEERING COLLEGE
AUTONOMOUS

Course Code: B23EEOE05					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
AUTONOMOUS VEHICLE SYSTEMS					
Offered by EEE					
(Offered to AIDS, AIML, CE, CIC, CSBS, CSE, CSG, CSIT, ECE, IT &ME)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer ONE Question from EACH UNIT					
All questions carry equal marks					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	What is a 2D rotation matrix and how is it used in autonomous vehicles?	1	2	2
	b).	Mention two key differences between control and decision-making modules in AVs.	1	2	2
	c).	What is the dynamic bicycle model	2	2	2
	d).	What are the states of an AV in the context of motion control?	2	2	2
	e).	What is brake-by-wire in autonomous vehicles?	3	2	2
	f).	Explain the working principle of LiDAR	3	2	2
	g).	Draw a simple FSM for basic driving behaviors	4	2	2
	h).	What is the difference between behavior planning and motion planning in autonomous vehicles?	4	2	2
	i).	What is a URDF file in ROS and why is it used?	5	2	2
	j).	Name any two commonly used real-time topics in ROS2 and their purpose.	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Describe the five levels of vehicle autonomy as defined by SAE. Highlight the differences in human involvement at each level.	1	3	5
	b).	Describe coordinate transformation between reference frames used in AVs	1	3	5
		OR			
3.	a).	Explain the basic architecture of an autonomous vehicle. Label the components under sensors, perception, planning, control, and decision-making modules.	1	3	5
	b).	How does forward kinematics affect vehicle path tracking, and what constraints are imposed by the vehicle's kinematic model?	1	4	5

		UNIT-2			
4.	a).	Describe path tracking and list the main control objectives involved in motion control of autonomous vehicles.	2	3	5
	b).	Illustrate PID control for lane-following in autonomous vehicles and analyze its feedback mechanism using a block-diagram.	2	4	5
		OR			
5.	a).	Explain the difference between longitudinal and lateral vehicle dynamics.	2	3	5
	b).	What are ride comfort metrics in autonomous vehicles, and how do they influence motion control strategies?	2	4	5
		UNIT-3			
6.	a).	Compare LiDAR, Radar, and Camera sensors in autonomous vehicles based on their range, image clarity, and ability to work in different weather conditions.	3	3	5
	b).	Describe the role of Kalman Filter in state estimation. Use a block diagram to explain its two main steps: prediction and update.	3	4	5
		OR			
7.	a).	Illustrate a basic sensor fusion setup using LiDAR and camera, and explain how combining their data improves object detection.	3	3	5
	b).	Describe the architecture of a SLAM system used in autonomous navigation.	3	4	5
		UNIT-4			
8.	a).	Explain the use of Finite State Machines (FSMs) in autonomous driving. Design a simple FSM to model different driving behaviors such as cruising, stopping, and turning at intersections.	4	3	5
	b).	Discuss the role of decision trees in maneuver selection in autonomous vehicles. Create a sample decision tree for choosing between lane change, maintain lane, or slow down actions based on traffic conditions.	4	4	5
		OR			
9.	a).	Describe the Markov Decision Process (MDP) framework used in planning for autonomous vehicles. Illustrate with an example how states, actions, transition probabilities, and rewards define a driving scenario.	4	4	5
	b).	Explain reward engineering in reinforcement learning for autonomous vehicles, emphasizing the role of safety, comfort, and efficiency metrics in shaping the reward function.	4	4	5
		UNIT-5			
10.	a).	Explain the architecture of ROS2 and describe how nodes communicate	5	3	5

		using topics, services, and actions. Include a simple diagram.			
	b).	Illustrate and explain the flow of data between sensors (LiDAR, camera, GPS) and ROS2 nodes. Use topics like /lidar/points, /camera/info, and /gps/fix in your explanation.	5	4	5
		OR			
11.	a).	Discuss the organization of ROS2 workspace and package management. Provide a step-by-step process for creating a custom package.	5	3	5
	b).	Describe the role of ROS_control and ROS2 driver interfaces in implementing a drive-by-wire system. Explain with a real-world example.	5	4	5

COURSE OUTCOME

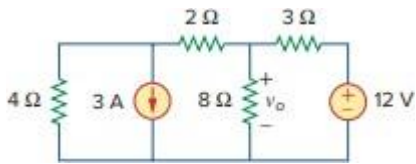
KL-KNOWLEDGE LEVEL

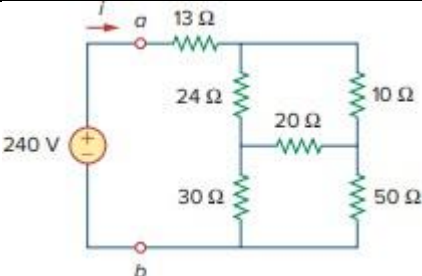
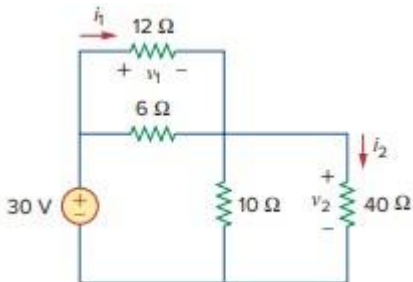
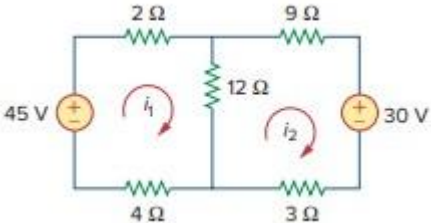
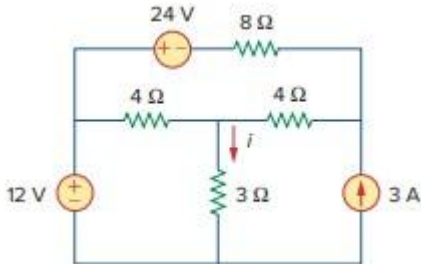
M-MARKS

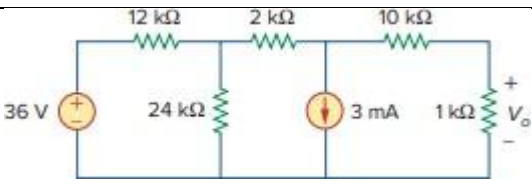
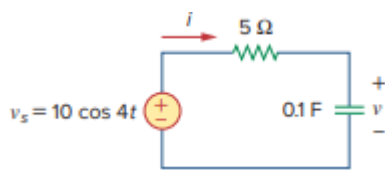
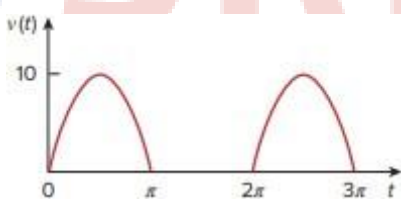
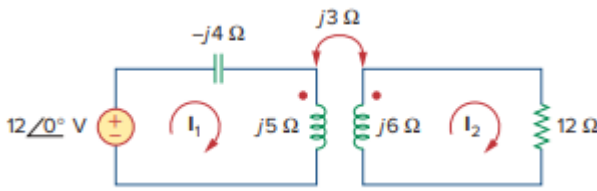
NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

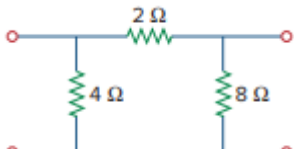


SRKR
ENGINEERING COLLEGE
AUTONOMOUS

Course Code: B23EEOE06					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
ELECTRICAL CIRCUIT ANALYSIS					
Offered by EEE					
(Offered to AIDS, AIML, CE, CIC, CSBS, CSE, CSG, CSIT, ECE, IT &ME)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Distinguish between active and passive element.	1	2	2
	b).	Indicate the relation between nodes, branches, and loops in an electric circuit.	1	2	2
	c).	Explain the principle of reciprocity.	2	2	2
	d).	Show the condition for maximum power when load resistance is equal to source resistance.	2	3	2
	e).	Represent the Phasor for the sinusoid $v = 20 \cos (2t+400) \text{ V}$.	3	2	2
	f).	Draw the power triangle.	3	2	2
	g).	Distinguish between self and mutual inductance.	4	2	2
	h).	Explain about balanced phase voltages in a three-phase system.	4	2	2
	i).	Sketch the frequency vs current plot in a series RLC circuit	5	3	2
	j).	Define Z-parameters.	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Calculate V_O using source transformation for the given circuit. 	1	4	5
	b).	Calculate equivalent resistance R_{ab} and find current 'i' in the circuit shown.	1	4	5

					
		OR			
3.	a).	Analyze the circuit shown to find i_1 and i_2 using current division rule. 	1	4	5
	b).	Calculate currents i_1 and i_2 in the circuit shown using mesh analysis. 	1	4	5
		UNIT-2			
4.	a).	State and Explain The venin's Theorem with an example.	2	3	5
	b).	Calculate current 'i' in the circuit shown using superposition theorem. 	2	4	5
		OR			
5.	a).	Calculate V_o in the given circuit using Norton's theorem.	2	4	5

					
	b).	State and Prove Maximum power transfer Theorem.	2	3	5
		UNIT-3			
6.	a).	Illustrate the relationship for Phasor voltage and current in an inductor And draw the Phasor diagram.	3	4	5
	b).	Calculate $v(t)$ and $i(t)$ in the circuit shown 	3	4	5
		OR			
7.	a).	Calculate the RMS value of the wave form shown 	3	4	5
	b).	The voltage across the load is $v(t)=60 \cos(\omega t-10^\circ)$ V and current through the element in the direction of voltage drop is $i(t)=1.5 \cos(\omega t+50^\circ)$ A Calculate (a) complex and apparent powers(b) real and reactive powers(c) load impedance.	3	4	5
		UNIT-4			
8.	a).	Calculate the Phasor currents I_1 and I_2 in the circuit shown 	4	4	5
	b).	Obtain the expression for coefficient of coupling and analyze how mutual inductance effects the coefficient of coupling?	4	4	5

		OR			
9.	a).	Analyze the relation between line voltage, phase voltage and line current and phase current in a star connection.	4	4	5
	b).	A balanced star connected load of $4+j3$ per phase is connected to a balanced 3-phase 400V supply. The phase current is 12A. Calculate (i) the total active power (ii) reactive power (iii) total apparent power.	4	4	5
		UNIT-5			
10.	a).	A series connected circuit has $R=4\Omega$, and $L=25\text{mH}$. Calculate the value of C that will produce a quality factor of 50. Find ω_1 , ω_2 and B.	5	4	5
	b).	Illustrate the expression for resonant frequency, band width, quality factor in a series resonance circuit.	5	4	5
		OR			
11.	a).	Determine h-Parameters for the given network 	5	4	5
	b).	Determine Y- Parameters in terms of Z- parameters.	5	4	5

COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Estd. 1980

AUTONOMOUS

Course Code:B23ITOE03					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
PRINCIPLES OF DATA BASE MANAGEMENT SYSTEMS					
Offered by IT					
(offered to CE, ECE, EEE & ME)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define DBMS.	1	2	2
	b).	List the different database models.	1	2	2
	c).	Explain the aggregation relationship with an example.	2	2	2
	d).	Define cardinality.	2	2	2
	e).	List the DDL and DML commands.	3	2	2
	f).	What is the difference between an updatable view and non-updatable view?	3	2	2
	g).	Define a functional dependency.	4	2	2
	h).	What is a well-structured relation in normalization?	4	2	2
	i).	Define a deadlock.	5	2	2
	j).	List the transaction states.	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Illustrate the database system structure with a neat diagram.	1	2	5
	b).	Present any two database applications by describing their features.	1	2	5
		OR			
3.	a).	Explain data independence and its types in detail.	1	2	5
	b).	Explain the role of a database administrator.	1	2	5
		UNIT-2			
4.	a).	Construct an ER diagram for the university registrar's office. The office maintains data about each class, including the instructor, the enrollment and the time and place of the class meetings. For each student class pair a grade is recorded. Determine the entities and relationships.	2	3	5

	b).	Describe the integrity constraints with examples.	2	2	5
		OR			
5.	a).	Explain the different types of attributes in an ER model.	2	2	5
	b).	Define the following terms and give examples. i) cardinality ii) unary relationships iii) aggregation iv) specialization.	2	2	5
		UNIT-3			
6.	a).	Explain the different data types in SQL.	3	2	5
	b).	Explain about different DML operations.	3	2	5
		OR			
7.	a).	Write SQL statements for following : Student(enrno, name courseid, emailid, cellno), course(courseid, course_nm, duration). i) Add a column city in the student table. ii) Find a list of students who have enrolled in a “computer” course. iii) List name of all courses with their duration. iv) List names of all students start with “a”. v) List emailid and cellno of all mechanical engineering students.	3	3	5
	b).	Explain the different types of joins with examples.	3	2	5
		UNIT-4			
8.	a).	Explain the functional dependency with multi-valued dependencies with examples.	4	2	5
	b).	Explain insertion, deletion and modification anomalies with suitable examples.	4	2	5
		OR			
9.	a).	State BCNF. How does it differ from 3NF?	4	2	5
	b).	How to compute closure of a set of functional dependencies? Explain with a suitable example schema.	4	2	5
		UNIT-5			
10.	a).	Explain ACID properties.	5	2	5
	b).	Explain 2-Phase Locking Protocol.	5	2	5
		OR			
11.	a).	Explain various deadlock prevention methods.	5	2	5
	b).	Describe serializability with an example.	5	2	5

COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23MEOE07																													
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23																								
IV B.Tech. I Semester MODEL QUESTION PAPER																													
TOTAL QUALITY MANAGEMENT																													
Offered by MECH																													
(Offered to AIDS, AIML, CE, CIC, CSBS, CSE, CSG, CSIT, ECE, EEE & IT)																													
Time: 3 Hrs.			Max. Marks: 70 M																										
Answer Question No.1 compulsorily																													
Answer ONE Question from EACH UNIT																													
Assume suitable data if necessary																													
10 x 2 = 20 Marks																													
			CO	KL	M																								
1.	a).	Define quality.	1	1	2																								
	b).	What are the dimensions of quality?	1	1	2																								
	c).	Differentiate between Process and customer.	2	1	2																								
	d).	Who is an internal customer?	2	1	2																								
	e).	What are the key reasons for the transition of traditional to TQM approach?	3	1	2																								
	f).	What is Quality Circle?	3	1	2																								
	g).	Give two examples of quality appraisal costs.	4	1	2																								
	h).	Give three components of external failure costs.	4	1	2																								
	i).	Are ISO 9000 standards, process based or product based?	5	1	2																								
	j).	What does ISO 9001:2000 signify?	5	1	2																								
5 x 10 = 50 Marks																													
		UNIT-1																											
2.		Construct the number of defect chart for the data given below:	1	3	10																								
	Sub group	1				2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
	No.of defect	2				3	0	1	3	5	3	1	2	0	1	0	2	4	1	2	0	3	2	1	4	0	0	3	
		Mark the central level, UCL and LCL and plot.																											
		OR																											
3.	a).	What is acceptance sampling? What are its advantages and disadvantages?	1	3	5																								
	b).	What are the benefits of TQM?	1	3	5																								
		UNIT-2																											

4.		Explain in brief how does marketing and sales plays crucial role in TQM?	2	3	10
		OR			
5.		Define benchmarking and Explain the process of benchmarking.	2	3	10
		UNIT-3			
6.		Explain briefly what is meant by quality circle.	3	3	10
		OR			
7.		What is the role of teams in organizing for TQM?	3	3	10
		UNIT-4			
8.		Discuss the various types of quality costs.	4	3	10
		OR			
9.		What is quality management? How do you measure Quality cost?	4	3	10
		UNIT-5			
10.	a).	Explain the benefits of ISO certification.	5	3	5
	b).	What is quality policy statement?	5	3	5
		OR			
11.		What are the various quality-documentation requirement for ISO 9000 certification?]	5	3	10

COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23MEOE08					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
SMART MANUFACTURING					
Offered by MECH					
(Offered to AIDS, AIML, CE, CIC, CSBS, CSE, CSG, CSIT, ECE, EEE & IT)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define circular manufacturing.	1	1	2
	b).	Differentiate Industry 4.0 and Industry 5.0.	1	2	2
	c).	Identify a smart sensor used for tool wear monitoring.	2	1	2
	d).	Draw the block diagram of a smart sensor interface used in IIoT.	2	1	2
	e).	Explain the Cognition level in 5C architecture.	3	2	2
	f).	Compare SCADA systems and Cyber-Physical Systems.	3	2	2
	g).	Explain the role of a Digital Twin in manufacturing.	4	2	2
	h).	Classify the Six Big Losses in manufacturing.	4	2	2
	i).	Explain the TCP/IP model used in industrial communication.	5	2	2
	j).	Compare MQTT and HTTP protocols.	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Explain the development from Industry 3.0 to Industry 4.0. How do smart machines support mass customization?	1	3	5
	b).	Explain what are the challenges do small industries face while implementing smart manufacturing? Suggest two solutions.	1	3	5
		OR			
3.	a).	Explain how circular economy concepts can be used in an automobile assembly plant. Mention three stages of resource optimization.	1	3	5
	b).	Explain the stages of Industry 4.0 evolution. At which stage does a factory move from visibility to predictability? Give an example.	1	3	5
		UNIT-2			
4.	a).	A factory faces downtime due to motor overheating. Design a sensor system to monitor temperature and vibration.	2	3	5
	b).	Explain the role of smart actuators in a smart manufacturing system.	2	3	5

		How are they different from traditional actuators?			
		OR			
5.	a).	Explain how old machines can be connected to an IIoT network using smart sensors.	2	3	5
	b).	Draw and explain the block diagram of an IIoT architecture from sensor to cloud.	2	3	5
		UNIT-3			
6.	a).	Explain the 5C architecture for a robotic welding system.	3	3	5
	b).	Compare traditional programmed manufacturing systems with Cyber-Physical autonomous systems.	3	3	5
		OR			
7.	a).	Explain how the Cyber level in 5C architecture uses Digital Twins for production planning.	3	3	5
	b).	Explain how Cyber-Physical Systems help in reducing the Six Big Losses.	3	3	5
		UNIT-4			
8.	a).	Explain the steps involved in creating a Digital Twin for a wind turbine to predict blade damage.	4	3	5
	b).	Differentiate Supervised and Unsupervised Machine Learning. Explain one method used for anomaly detection in conveyor systems.	4	3	5
		OR			
9.	a).	Develop a predictive maintenance method for a hydraulic press using AI/ML. Which Six Big Loss is reduced by this method?	4	3	5
4	3	Explain the concept of Multiverse in Digital Twins with AR/VR for remote maintenance.			5
		UNIT-5			
10.	a).	Explain the requirements of a mesh network in a smart warehouse. Why is it better than star topology?	5	3	5
	b).	Explain the use of MQTT or CoAP protocols when many sensors send data with low congestion.	5	3	5
		OR			
11.	a).	Explain the importance of Physical Layer IoT protocols in high-speed motion control systems.	5	3	5
	b).	Draw the TCP/IP model and explain how safe transmission of an Emergency Stop signal is achieved in a factory network.	5	3	5

COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23BSOE03					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
MATHEMATICAL MODELLING FOR DATA SCIENCE					
Offered by E&H					
(Offered to AIDS, AIML, CE, CIC, CSBS, CSE, CSG, CSIT, ECE, EEE, IT & ME)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a)	Define Joint Probability distribution function for discrete random variables.	1	1	2
	b)	Define Stochastic Matrix and write its properties	1	2	2
	c)	Explain Queuing theory with block diagram	2	2	2
	d)	Explain the concept of ‘Networks of queues’.	2	2	2
	e)	Describe Monte-Carlo simulation	3	1	2
	f)	State the Phases of Simulation Models	3	1	2
	g)	What are the causes of variation? Define.	4	1	2
	h)	Define ANOVA, what are its assumptions?	4	1	2
	i)	Distinguish between parametric and non- parametric methods	5	1	2
	j)	Write the advantages and disadvantages of non-parametric methods.	5	1	2
5 x 10 = 50Marks					
		UNIT- I			
2.	a)	Every year, a man trades his car for a new car. If he has a Maruti, he trades it for an Ambassador. If he has an ambassador, he trades it for Santro. However, if he has a Santro, he is just as likely to trade it for a new Santro as to trade it for a Maruti or an Ambassador. In 2020 he bought his first car, which was a Santro. (a) Calculate the probability that he has (i) 2022 Santro (ii) 2022 Maruti (iii) 2023 Ambassador (iv) 2023 Santro.	1	3	10
		OR			
3.	a)	Compute (a) marginal distributions $f(x)$ and $g(y)$ (b) $E(X)$ and $E(Y)$ (c) $Cov(X,Y)$ (d) σ_x and σ_y and (e) $\rho(X,Y)$ for the following bivariate distribution	1	3	10

		<table><tr><td>x \ Y</td><td>-4</td><td>2</td><td>7</td></tr><tr><td>1</td><td>1/8</td><td>1/4</td><td>1/8</td></tr><tr><td>5</td><td>1/4</td><td>1/8</td><td>1/8</td></tr></table>	x \ Y	-4	2	7	1	1/8	1/4	1/8	5	1/4	1/8	1/8											
x \ Y	-4	2	7																						
1	1/8	1/4	1/8																						
5	1/4	1/8	1/8																						
		UNIT- II																							
4.	a)	A T.V. repairman finds that the time spent on his jobs has an exponential distribution with mean 30 minutes. He repairs sets in the order in which they arrive. The arrival of the sets is approximately Poisson with an average of 10 per an eight-hour day. Calculate the repairman’s idle time each day. How many jobs are ahead of the average set just brought in?	2	3	10																				
		OR																							
5	a)	In a railway marshalling yard, goods trains arrive at a rate of 30 trains per day. Assuming that inter-arrival times follow an exponential distribution and the service time distribution is also exponential with an average 36 minutes. Calculate the following: (i) The average number of trains in the queue. (ii) The Probability that the queue size exceeds 10 If the input trains increases to average 33 per day, what will be the change in (i) & (ii)?	2	3	10																				
		UNIT- III																							
6.	a)	An automobile company manufactures around 150 scooters. The daily production varies from 146 to 154 depending upon the availability of raw material and other working conditions: <table><tr><td>Production (per day)</td><td>146</td><td>147</td><td>148</td><td>149</td><td>150</td><td>151</td><td>152</td><td>153</td><td>154</td></tr><tr><td>Probability</td><td>0.04</td><td>0.09</td><td>0.12</td><td>0.14</td><td>0.11</td><td>0.20</td><td>0.20</td><td>0.12</td><td>0.08</td></tr></table> The finished scooters are transported in a specially arranged lorry accommodating 150 scooters. Using the following random numbers: 80 91 76 75 64 43 18 26 10 12 65 68 69 61 57 Simulate the process to compute (a) The average number of scooters waiting in the factory. (b) The average number of empty spaces on the lorry.	Production (per day)	146	147	148	149	150	151	152	153	154	Probability	0.04	0.09	0.12	0.14	0.11	0.20	0.20	0.12	0.08	3	3	10
Production (per day)	146	147	148	149	150	151	152	153	154																
Probability	0.04	0.09	0.12	0.14	0.11	0.20	0.20	0.12	0.08																
		OR																							
7.	a)	Describe what simulation? Explain the steps in simulation process with a flow chart.	3	2	10																				
		UNIT- IV																							
8.	a)	Explain the inferential procedure in “Analysis of Variance Two-way classification”	4	3	10																				

		OR																																											
9.	a)	As head of the department of a consumer’s research organization, you have the responsibility for testing and comparing lifetimes of four brands of electric bulbs. The data are shown below, each entry represents the lifetime (in hundreds of hours of an electric bulb. <table><tr><th colspan="4">Brand</th></tr><tr><th>A</th><th>B</th><th>C</th><th>D</th></tr><tr><td>20</td><td>25</td><td>24</td><td>23</td></tr><tr><td>19</td><td>23</td><td>20</td><td>20</td></tr><tr><td>21</td><td>21</td><td>22</td><td>20</td></tr></table> Can we interpret that the mean lifetimes of the four brands of electric bulbs are equal?	Brand				A	B	C	D	20	25	24	23	19	23	20	20	21	21	22	20	4	3	10																				
Brand																																													
A	B	C	D																																										
20	25	24	23																																										
19	23	20	20																																										
21	21	22	20																																										
		UNIT- V																																											
10.		A coin is tossed 20 times and the following sequence of heads (H) and tails (T) are obtained. H T T H H T T H T H H H T T H H T H H H Apply Run test to determine at 5% level of significance weather the coin is unbiased.	5	3	10																																								
		OR																																											
11.		Describe the test procedure of Kendall’s test. The ranks of 9 students of a class in four subjects: English, Mathematics, Economics and Computer science are as follows. Develop Kendall Concordance test at 5% level of significance. Clearly state the null hypothesis for the test. <table><tr><td>English</td><td>4</td><td>7</td><td>9</td><td>3</td><td>8</td><td>2</td><td>6</td><td>5</td><td>1</td></tr><tr><td>Mathematics</td><td>3</td><td>6</td><td>8</td><td>1</td><td>7</td><td>5</td><td>9</td><td>2</td><td>4</td></tr><tr><td>Economics</td><td>5</td><td>7</td><td>6</td><td>1</td><td>9</td><td>2</td><td>8</td><td>3</td><td>4</td></tr><tr><td>Computers</td><td>4</td><td>8</td><td>7</td><td>2</td><td>6</td><td>1</td><td>9</td><td>5</td><td>3</td></tr></table>	English	4	7	9	3	8	2	6	5	1	Mathematics	3	6	8	1	7	5	9	2	4	Economics	5	7	6	1	9	2	8	3	4	Computers	4	8	7	2	6	1	9	5	3	5	3	10
English	4	7	9	3	8	2	6	5	1																																				
Mathematics	3	6	8	1	7	5	9	2	4																																				
Economics	5	7	6	1	9	2	8	3	4																																				
Computers	4	8	7	2	6	1	9	5	3																																				

COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks